223

A Physics-Based Three-Dimensional Model for Heat Transfer and Stress Analysis in Metal Additive Manufacturing

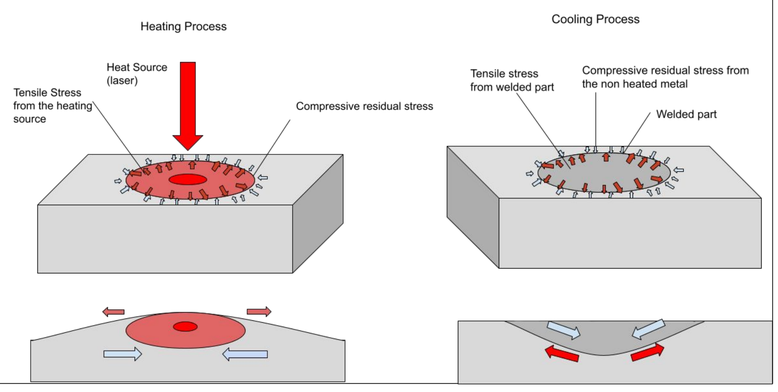
Kamryx Davis and Milton Tomas
Department of Electrical and Computer Engineering

Faculty Mentor: Dr. Hayri Sezer

Introduction

Residual Stress

- Unintentional residual stress in metals can lead to catastrophic events such as the Silver Bridge collapse. Residual stress accumulated during the metal additive manufacturing process can lead to part and system failure. By reducing residual stress we can increase part integrity and potentially save lives.



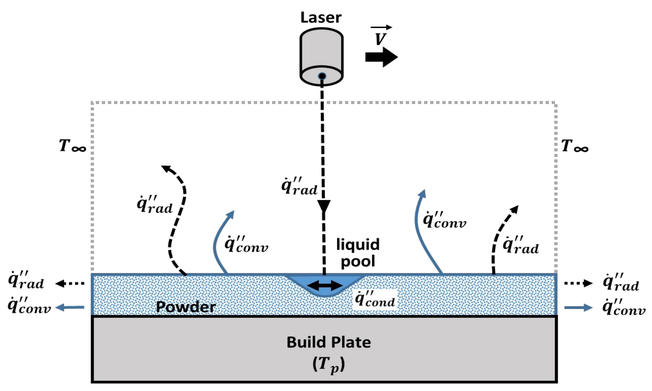
Preliminary Work

- This study investigates residual stress in laser powder bed fusion (LPBF) / direct metal laser sintering (DMLS), emphasizing the effect of scan patterns on thermal gradients and stress development. A 3D thermal-stress model was developed, incorporating realistic thermal properties and phase change. Among several patterns, a novel **Hexagonal Zigzag** with **non-linear island intersections** showed potential to reduce residual stress by promoting more uniform heat distribution. These initial findings support future work on scan path optimization and high-performance simulation.
- From conducting a literature review, scan pattern characteristics such as the number of turns, scan pass length, and rotation affect residual stress.

Mathematical Model

- Temperature-dependent thermophysical properties
- Solid-liquid phase change using enthalpy method
- Gaussian distribution of power source
- Stress is calculated after solidification process takes place, not in the melt pool
- Cell centered finite volume method is used for the discretization of the partial differential equations (PDEs)

Laser Powder Bed (LPBF) Process



Thermal Model

$$Q = -\frac{\partial(\rho \Delta H)}{\partial t} \qquad \rho C_p \frac{\partial T}{\partial t} = \nabla.(K \nabla T) + Q$$

$$\Delta H = C_p T + L l_f^{n-1} \qquad q''_{conv} = h(T_\infty - T_s)$$

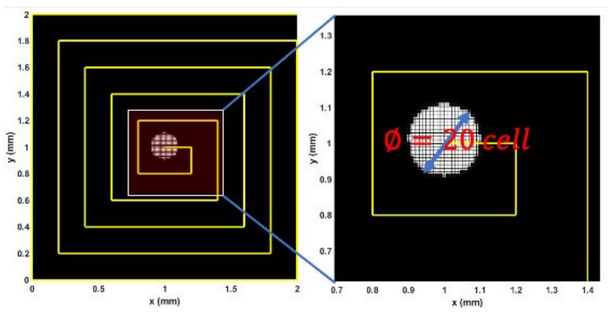
$$q''_{rad} = \sigma \epsilon (T_\infty^4 - T_s^4)$$

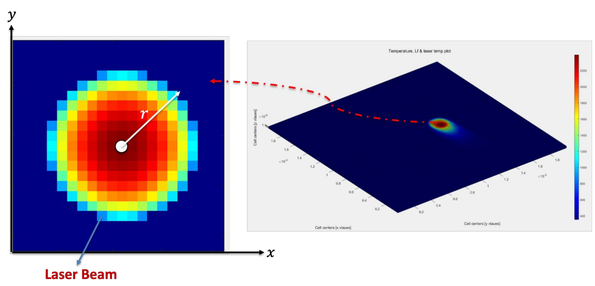
Stress Analysis

$$\{\sigma\} = [D]\{\varepsilon^e\} \qquad \varepsilon^{th} = \int_{T_{ref}}^{T} \alpha_e(T) dT$$

$$\sigma_v = \frac{1}{\sqrt{2}} \sqrt{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2}$$

Time, Length Scales, and Power Source





$\emptyset = 200\ \mu m \rightarrow 20$ computational cell

$$dx = \frac{200\mu m}{20} = 10\mu m$$

Domain: $3 \times 3\ mm \rightarrow 300 \times 300 = 90000\ cells$
At each layer, we take 3 computational cell
For one layer 270000 cells
For 3 layers: 810000 computational cells

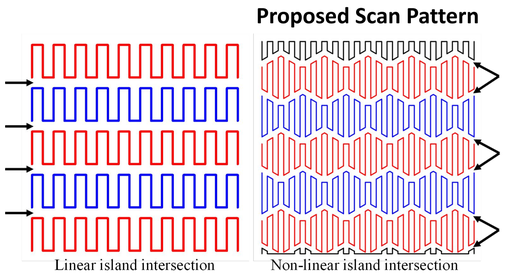
$$q''_{sph} = \frac{3P}{2\pi r^2} \sqrt{\left|1 - \frac{(x-x_0)^2}{r^2} - \frac{(y-y_0)^2}{r^2}\right|} \; for\ [x - x_0]$$

Scanning velocity $v = 2m/s$

$$dt = \frac{dx}{v} = \frac{10\mu m}{2m/s} = 5 \times 10^{-6} s.$$

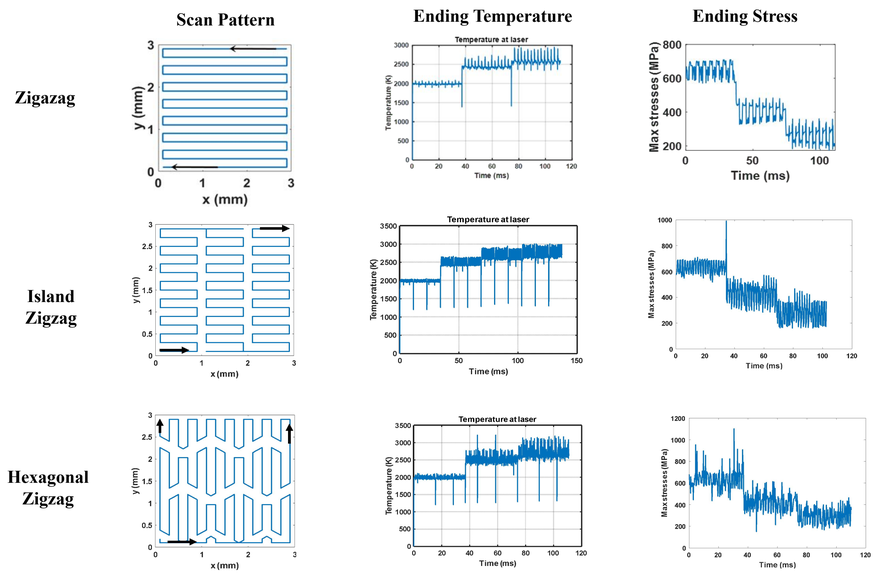
Approaches

- Utilize a cell-centered finite volume method implemented in MATLAB with a fully implicit time integration scheme
- Model features include a layer-by-layer approach, accommodating a growing computational domain as new material is deposited
- Model integrates various scanning patterns, such as inward and island-type rectangular strategies, enabling exploration of their impact on thermal gradients and final part properties
- Model incorporates a pseudo-steady state stress analysis at each time step

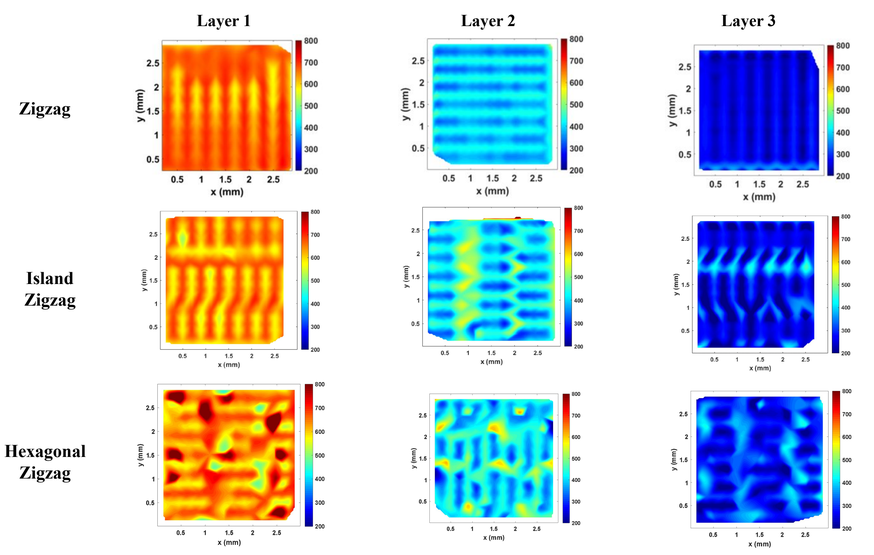


Results and Findings

Thermal Gradient Induced Residual Stress



Layer-wise Residual Stress



Discussion and Future Work

- This research provides a robust framework for predicting thermal histories and stress distributions during metal additive manufacturing, offering valuable guidance for optimizing process parameters and scanning strategies to achieve desired mechanical properties and minimize defects.
- Future work will:
  - Expand the computational domain to capture more realistic part geometries
  - Change scan strategies, including cell shape, scan direction, and pass overlaps
  - Conduct parametric studies to optimize laser power, scan speed, and layer thickness for further stress reduction
  - Implement GPU parallelization using Julia

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